

Unified Master Schema — Big Data Contacts

1. Master Table: `contacts_master`

Column	Type	Description
id	BIGINT	Unique record ID
person_name	TEXT	Full name of the person
email	TEXT	Email address
phone	TEXT	Phone number
city	TEXT	City name
state	TEXT	State/Province
country	TEXT	Country
company	TEXT	Company or Organization
job_title	TEXT	Job title or designation

2. Schema Normalization Rules

All incoming files (CSV, Excel, JSON, TSV) are mapped into a **unified schema**.

Input Field Examples:

- `Name, FullName, person, contact_name` → `person_name`
- `E-mail, mail, emailAddress` → `email`
- `mobile, contact_number, telephone` → `phone`

Standardized Output:

`person_name email phone city state country company job_title`

3. Duplicate Removal Strategy

Duplicate rows are detected using:

If both email & phone are NULL → record is kept (marked low confidence).

4. Supported Input Formats

Format	Method Used
CSV	<code>pandas.read_csv</code>

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TSV	pandas.read_csv (sep="\t")
Excel	pandas.read_excel
JSON	json module / pandas
JSONL	line-by-line parser
SQL dump	custom import + regex parsing

5. Example of Unified Record

```
{
  "id": 1636215,
  "person_name": "John Doe",
  "email": "john.doe@example.com",
  "phone": "+1-202-555-1234",
  "city": "New York",
  "state": "NY",
  "country": "USA",
  "company": "Example Corp",
  "job_title": "Project Manager"
}
```

6. Output Pipeline Stages

Stage	Description
Raw Input	Original file from dataset
Normalized	Unified field mapping completed
Cleaned	Bad rows fixed/removed
Deduped	Duplicate records removed
Final Output	Inserted into `contacts_master`

```
CREATE INDEX idx_country ON contacts_master(country);
CREATE INDEX idx_email ON contacts_master(email);
CREATE INDEX idx_phone ON contacts_master(phone);
CREATE INDEX idx_company ON contacts_master(company);
```

8. Future Schema Extensions

hashed_email, hashed_phone (privacy)

linkedin_url, domain, industry

enrichments from 3rd party APIs

 **When pasted, run:**

```
```bash
git add docs/schema/schema-definition.md
```

```
git commit -m "Add schema-definition.md"
git push
```